

TOP DATA ENGINEER SYNTAX INTERVIEW QUESTIONS (Most Asked)



1. SQL INTERVIEW QUESTIONS WITH ANSWERS

Q1. How do you find the 2nd highest salary?

Answer:

```
SELECT MAX(salary)
FROM employees
WHERE salary < (SELECT MAX(salary) FROM
employees);
```

Q2. How do you calculate running total using window functions?

Answer:

```
SELECT emp_id, amount,
       SUM(amount) OVER (PARTITION BY emp_id ORDER
BY txn_date) AS running_total
```

FROM transactions;

Q3. How to delete duplicates and keep the latest record?

Answer:

```
SELECT *
FROM (
    SELECT *,
           ROW_NUMBER() OVER (PARTITION BY id
                              ORDER BY updated_at DESC) AS rn
    FROM employee
) t
WHERE rn = 1;
```

Q4. Employees who joined in last 30 days

Answer:

```
SELECT *
FROM employees
WHERE join_date >= CURRENT_DATE - INTERVAL '30'
DAY;
```

Q5. How to pivot in SQL?

Answer:

```
SELECT emp_id,
       SUM(CASE WHEN month='Jan' THEN sales
              END) AS jan_sales,
       SUM(CASE WHEN month='Feb' THEN sales
              END) AS feb_sales
FROM sales_table
GROUP BY emp_id;
```

2. PYSPARK INTERVIEW QUESTIONS WITH ANSWERS

Q6. How to drop duplicates on specific columns?

Answer:

```
df = df.dropDuplicates(["id", "date"])
```

Q7. Write syntax for a PySpark window function.

Answer:

```
from pyspark.sql.window import Window
from pyspark.sql.functions import row_number

windowSpec =
Window.partitionBy("id").orderBy("timestamp")
df2 = df.withColumn("rn",
row_number().over(windowSpec))
```

Q8. How do you explode an array column?

Answer:

```
from pyspark.sql.functions import explode

df2 = df.withColumn("item", explode("items"))
```

Q9. How to write a PySpark DataFrame to Delta format with schema merge?

Answer:

```
df.write.format("delta").option("mergeSchema",
"true").mode("append").save(path)
```

Q10. How do you read a JSON file in PySpark?

Answer:

```
df = spark.read.json("/path/file.json")
```

3. PYTHON QUESTIONS WITH SHORT SYNTAX ANSWERS

Q11. What is dictionary comprehension?

Answer:

```
result = {k: v*2 for k, v in data.items()}
```

Q12. Lambda + filter example

Answer:

```
even_numbers = list(filter(lambda x: x % 2 == 0, numbers))
```

Q13. Exception handling syntax

Answer:

```
try:
    x = 10 / 0
except ZeroDivisionError:
    print("Error")
finally:
    print("Done")
```

4. AZURE DATA FACTORY (ADF) QUESTIONS WITH ANSWERS

Q14. Expression to get yesterday's date in ADF

Answer:

```
@{formatDateTime(addDays(utcNow(), -1), 'yyyy-MM-dd')}
```

Q15. ADF If Condition syntax

Answer:

```
@{if(equals(variables('mode'), 'incremental'),  
'INC', 'FULL')}
```

Q16. ADF Dynamic File Path

Answer:

```
@concat('raw/', pipeline().parameters.fileName,  
'/')
```

Q17. ADF parameter declaration

Answer:

```
"parameters": {  
  "fileName": {  
    "type": "string"  
  }  
}
```

5. SPARK SQL QUESTIONS WITH ANSWERS

Q18. How do you create a temp view in Spark?

Answer:

```
df.createOrReplaceTempView("emp_view")
```

Q19. How do you perform MERGE in Delta Lake?

Answer:

```
MERGE INTO target t  
USING source s  
ON t.id = s.id  
WHEN MATCHED THEN UPDATE SET *  
WHEN NOT MATCHED THEN INSERT *
```

Q20. How to cast a column in Spark SQL?

Answer:

```
SELECT CAST(salary AS INT) AS salary_int FROM  
employee;
```

6. ETL / PIPELINE QUESTIONS WITH ANSWERS

Q21. Basic Airflow DAG syntax

Answer:

```
from airflow import DAG  
from airflow.operators.bash import BashOperator  
  
with DAG('demo', schedule_interval='@daily') as  
dag:  
    t1 = BashOperator(task_id='run',  
bash_command='echo hello')
```

Q22. YAML Pipeline trigger syntax

Answer:

```
trigger:  
  branches:  
    include:  
      - main
```